Domain Addendum — Lecture Two

Study handout — text extracted from the original slides (wording preserved). Logic symbols fixed (∧ ∨ ¬ ⇒ ⇐ ≡ ≠ ⊥). About half the paper or less. 12 slides. Black text on white.

How to use: skim slide titles, study bullets in order, read formulas as written.

Slide 1 — Addendum to Number Theory

(No body text on this slide.)

Slide 2

Natural Numbers Natural numbers count the elements of a whole, cardinal number, or express the order which occupies an element in a whole, ordinal number.
• The set of the natural numbers is formed by:
nat = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9,...}
• 6 / 2 ∈ nat
• 2 / 6 ∉ nat
• We may use powers, since it is the abbreviated form of writing a product formed by several equal factors.
• The root of natural numbers is not always a natural number. This only occurs when the root is exact.

Slide 3 — Integers Integers are of the type:

int = {...−5, −4, −3, −2, −1, 0, 1, 2, 3, 4, 5...}
• We can operate with powers, but the exponent has to be a natural number.
The root of an integer is not always another integer.
This only occurs when the root is exact, of even index and positive radicand.

Slide 4 — Rational Numbers

• Arational number is the quotient of two integers with a nonzero denominator. rat □
rat= □ , □∈ □ □ □ , □ ≠ 0 □
• Non -recurrent decimal numbers are not rational numbers, but
• exact, repeating decimal numbers are.
Example: 22/7= 3.142857 142857 142857

Slide 5 — Rational Numbers

• The sum, difference, product and quotient of two rational numbers is another rational number.
• With rational numbers, we can operate with powers, but the exponent must be an integer.
• The root of a rational number is not always another rational number. This only happens when the root is exact, the index is even and the radicand is positive.
− ∉ □ □ □

Slide 6 — Irrational numbers

• Anumber is irrational if there are infinite, non-recurrent decimal places, therefore, it cannot be expressed in the form of fraction.
• The best known irrational number is
π, which is defined as the relationship between the perimeter of a circle and its diameter.
π = 3.141592653589...
• Other irrational number is The number, e, which appears in growth processes, in radioactive decay and ….
e = 2.718281828459...
• The golden ratio, a geometric proportion used by artists from all periods 1 + 5 □ = = 1.61803398874989…

Slide 7 — Real Numbers

• The set formed by rational and irrational numbers is the set of real numbers denoted by Real
• All operations can be performed with real numbers with the exception:
• The root of an even index and negative radicand, and
• Division by zero.

Slide 8 — irrational real rat int nat

(No body text on this slide.)

Slide 9 — The Real Line

• For every real number there is a point on a straight line and for every point on the straight line, a real number.

Slide 10

Imaginary Numbers An imaginary number is denoted by bi, where:
• b is a real number.
• i is the imaginary unit:
• With imaginary numbers, roots with an even index and negative roots can be calculated:
x + 9 = 0

Slide 11 — Complex Numbers

• Acomplex number is a binomial a + bi.
• a is the real part of the complex number.
• b is the imaginary part of the complex number.
• If b = 0, the complex number is reduced to a real number as a + 0 x i = a.
• If a = 0, the complex number is reduced to bi, and it is said that it is a pure imaginary number.
• The set of all complex numbers is denoted by C □ = □ + □ □ □ , □ : □ □ □ □

Slide 12

• The Number Theory - Back to Lecture 2